

C++ Programming Curriculum



COURSE CURRICULUM

C++ Programming Curriculum

Goal: Learn fundamental concepts of C++ with easy explanations and hands-on examples, enabling students to write simple programs and small applications confidently.

Week 1 – Getting Started

Day 1 – Introduction to Programming & C++

- What is programming?
- What is C++? Why use C++?
- Setting up a C++ environment (Code::Blocks, Visual Studio, or online compiler).
- First simple program

```
#include <iostream>
using namespace std;
```

```
int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

Day 2 – Basics of Syntax

- #include, main() function, return 0
- Comments: // and /* */
- cout and cin
- Example: Ask user's name and print greeting

Day 3 – Variables & Data Types

- int, float, double, char, string, bool
- Declaring variables
- Simple example: Store and print age, height, name

Day 4 – Arithmetic Operators

- +, -, *, /, %
- Simple calculator example

COURSE CURRICULUM

Day 5 – Simple Project: Interactive Calculator

- Take two numbers and operator (+, -, *, /)
- Perform calculation and display result

Week 2 – Control Flow

Day 6 – Conditional Statements

- if, if-else, else if
- Example: Check if number is positive, negative, or zero

Day 7 – Comparison & Logical Operators

- ==, !=, >, <, >=, <=
- &&, ||, !
- Example: Age eligibility check (e.g., for voting)

Day 8 – Loops: for, while, do-while

- for loop: print numbers 1 to 10
- while loop: repeat until user enters 0
- do-while loop example

Day 9 – Simple Project: Number Guessing Game

- Generate random number (use `rand() % 10 + 1`)
- User guesses and program gives hints (high/low)

Day 10 – Break and Continue

- When to use break and continue
- Example: Loop to print even numbers up to 20

COURSE CURRICULUM

Week 3 – Functions & Arrays

Day 11 – Functions in C++

- What is a function and why use it
- Syntax of function
- Example: Function to add two numbers

Day 12 – Arrays Basics

- Declare an array
- Store multiple values
- Example: Store 5 test scores

Day 13 – Array Operations

- Print array elements
- Find maximum number in array
- Calculate sum of array elements

Day 14 – Simple Project: Student Score Program

- Input scores of 5 students
- Display average and highest score

Day 15 – Revision + Mini Quiz

- Multiple choice questions
- Short coding exercises

Week 4 – Advanced Basics

Day 16 – Strings in C++

- string data type
- String input and output using cin, getline(), cout

COURSE CURRICULUM

Day 17 – String Functions

- `length()`, `substr()`, `find()`, `replace()`
- Example: Check if password matches preset password

Day 18 – Pointers (Introduction)

- What is a pointer
- Pointer syntax and simple example
- Example:

```
int a = 5;  
int *ptr = &a;  
cout << *ptr; // Outputs 5
```

Day 19 – Simple Project: Swap Two Numbers using Pointers

- Swap function using pointers

Day 20 – Final Project: Simple Menu-Driven App

- Menu example:
 1. Add Student Score
 2. View Scores
 3. Exit

Bonus Topic – File Handling Basics

Day 21 – Introduction to File Handling

- What is file handling?
- Why use file handling?
- Types of files: Text files (reading/writing)
- Difference between text files and binary files
- What are file modes in programming (read, write, append, etc.)?
- How to open and close a file safely?

COURSE CURRICULUM

Day 22 – Writing to a File

- Example:

```
#include <fstream>
using namespace std;

int main() {
    ofstream outFile("data.txt");
    outFile << "Hello, File!" << endl;
    outFile.close();
    return 0;
}
```

Day 23 – Reading from a File

- Example:

```
#include <fstream>
#include <iostream>
using namespace std;

int main() {
    ifstream inFile("data.txt");
    string line;
    while (getline(inFile, line)) {
        cout << line << endl;
    }
    inFile.close();
    return 0;
}
```

Day 24 – Simple Project: Save and Read Student Scores

- Let user enter scores, save to file
- Read and display scores from file

Final Outcome

By the end of this curriculum, students will be able to:

- Write simple C++ programs
- Use variables, conditions, loops, functions, arrays, pointers, and file handling
- Build small interactive applications
- Perform basic string manipulation and file read/write



Scan the QR to know more